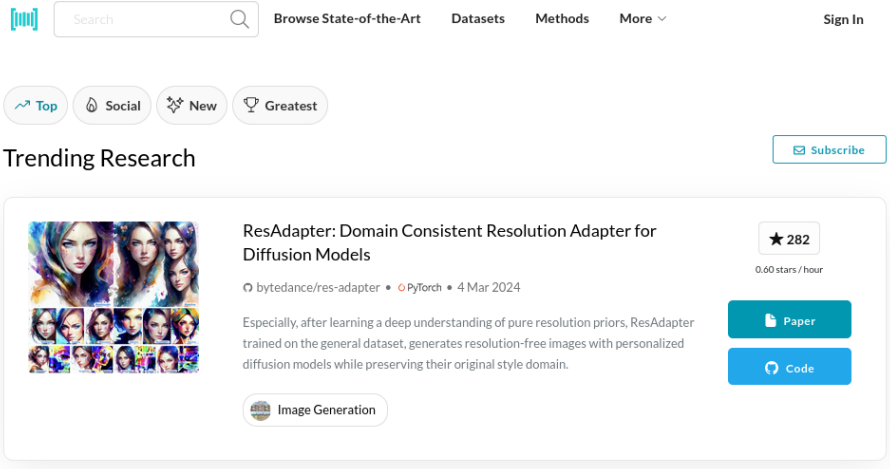
HyperPIE: Hyperparameter Information Extraction from Scientific Publications

Tarek Saier, Mayumi Ohta, Takuto Asakura, Michael Färber

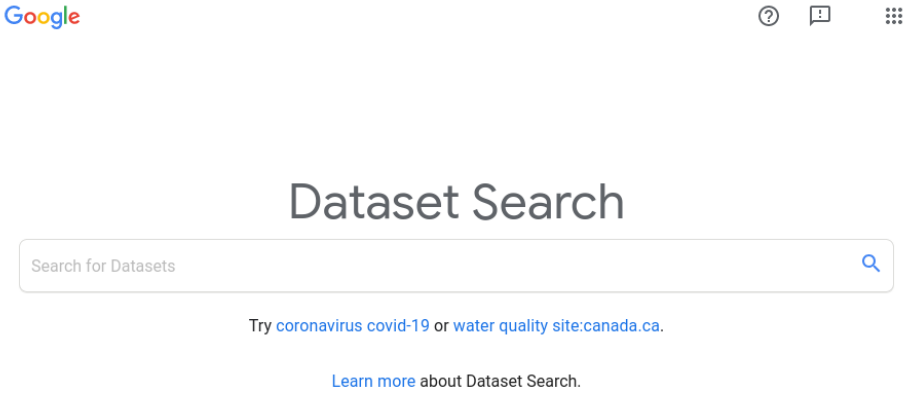
Tarek Saier | 27. March 2024



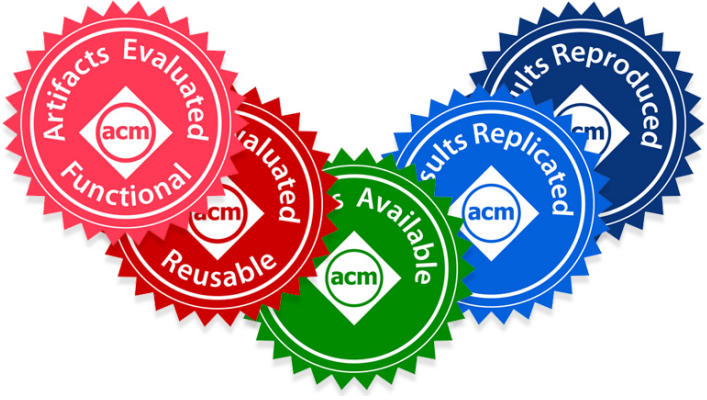
Motivation


Search
Browse State-of-the-Art
Datasets
Methods
More
Sign In
Top
Social
New
Greatest
Trending Research
Subscribe
ResAdapter: Domain Consistent Resolution Adapter for Diffusion Models
bytedance/res-adapter • PyTorch • 4 Mar 2024
Especially, after learning a deep understanding of pure resolution priors, ResAdapter trained on the general dataset, generates resolution-free images with personalized diffusion models while preserving their original style domain.
Image Generation
★ 282
0.60 stars / hour
Paper
Code

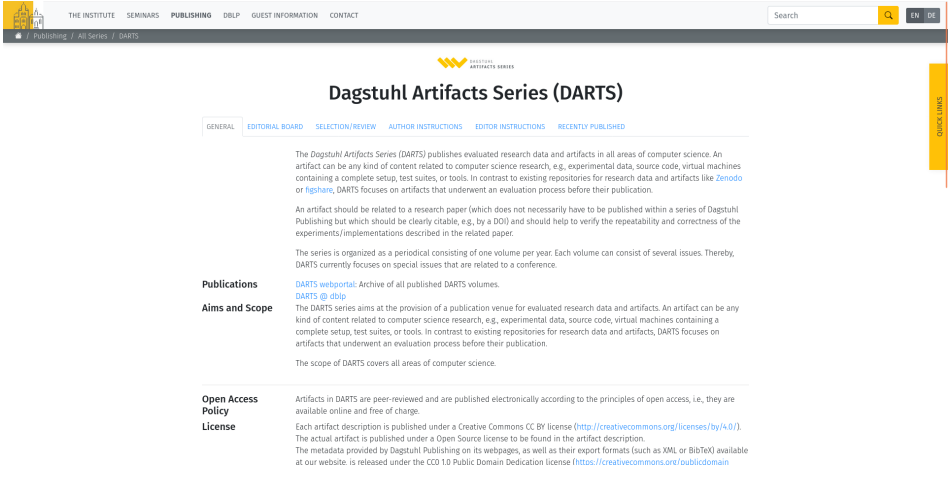
Motivation


Google
Dataset Search
Search for Datasets
Try coronavirus covid-19 or water quality site:canada.ca.
Learn more about Dataset Search.

Motivation


Artifacts Evaluated
acm
Functional
Reusable
Available
Results Replicated
Results Reproduced

Motivation


THE INSTITUTE
SEMINARS
PUBLISHING
DBLP
GUEST INFORMATION
CONTACT
Search
EN
DE
/ Publishing / All Series / DARTS
Dagstuhl Artifacts Series (DARTS)
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AUTHOR INSTRUCTIONS
EDITOR INSTRUCTIONS
RECENTLY PUBLISHED
QUICK LINKS
The Dagstuhl Artifacts Series (DARTS) publishes evaluated research data and artifacts in all areas of computer science. An artifact can be any kind of content related to computer science research, e.g., experimental data, source code, virtual machines containing a complete setup, test suites, or tools. In contrast to existing repositories for research data and artifacts like Zenodo or figshare, DARTS focuses on artifacts that underwent an evaluation process before their publication.
An artifact should be related to a research paper (which does not necessarily have to be published within a series of Dagstuhl Publishing but which should be clearly citable, e.g., by a DOI) and should help to verify the repeatability and correctness of the experiments/implementations described in the related paper.
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Publications
DARTS webportal: Archive of all published DARTS volumes.
DARTS @ dblp
Aims and Scope
The DARTS series aims at the provision of a publication venue for evaluated research data and artifacts. An artifact can be any kind of content related to computer science research, e.g., experimental data, source code, virtual machines containing a complete setup, test suites, or tools. In contrast to existing repositories for research data and artifacts, DARTS focuses on artifacts that underwent an evaluation process before their publication.
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Motivation

NLP field increasing focus on **data** and its **algorithmic processing**

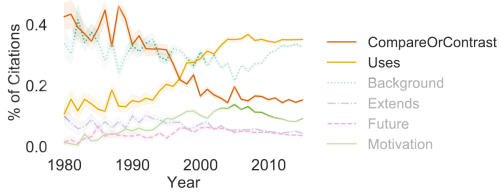


Figure 5: Changes in the average citation frame in ACL papers reveals a continued decline in the percentage of COMPARISON OR CONTRAST and increase in USES citations. The increase in BACKGROUND citations circa 2010 marks the start of the era of unlimited references in ACL conferences. Shaded regions show bootstrapped 95% confidence intervals.

Jurgens, D. *et al.* Measuring the Evolution of a Scientific Field through Citation Frames. *Transactions of the Association for Computational Linguistics* **6,** 391–406 (2018)

Hyperparameter Information Extraction from Scientific Publications

Hyperparameter Information Extraction from Scientific Publications

Hyperparameter Information Extraction from Scientific Publications

Task

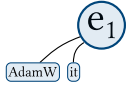
AdamW is chosen as the optimizer. We use it with a learning rate (α) of 1e-3 for all non BERT weights. For BERT weights we set α to 2e-5.

text example
based on arXiv:2005.00512

Task

AdamW is chosen as the optimizer. We use it with a learning rate (α) of 1e-3 for all non BERT weights. For BERT weights we set α to 2e-5.

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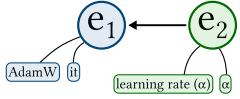


entity type
artifact

Task

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based on arXiv:2005.00512

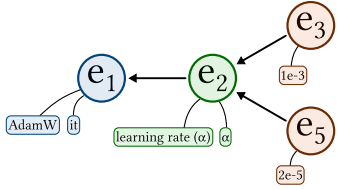


entity type
artifact parameter

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text example
based on arXiv:2005.00512

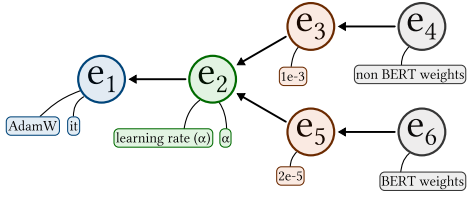


entity type
artifact parameter value

Task

AdamW is chosen as the optimizer. We use it with a learning rate (α) of 1e-3 for all non BERT weights. For BERT weights we set α to 2e-5.

text example
based on arXiv:2005.00512



entity type
artifact parameter value context

Overview

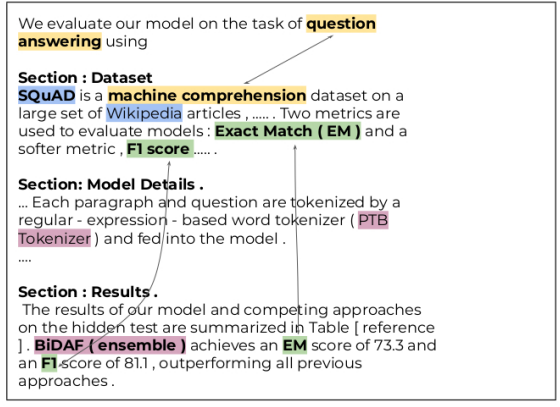
- **Related Work**
  - Fine-Tuned Models
  - LLMs
- **Approach**
  - Fine-Tuned Models
  - LLMs
- **Data**
- **Experiments**
  - Fine-Tuned Models
  - LLMs
- **Application**
- **Discussion**

Related Work: Fine-tuned models

- SciERC dataset
  - SCIIE [2]
  - PL-Marker [3]
  - → **entity type overlap**

[...]

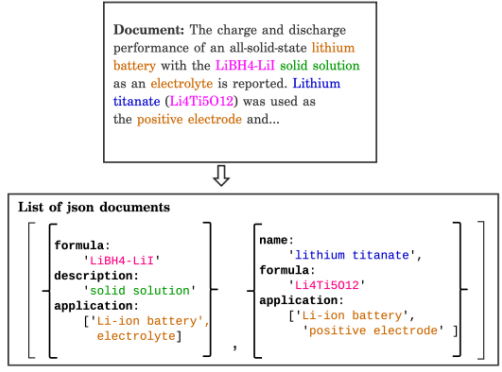
- SemEval 2021 (measurements)
  - LIORI [7]
  - → **utilize mention pattern regularities**



Related Work: LLMs

- Medical science [8]
  - singular values
  - lists
- Material science [9–11]
  - singular values
  - lists
  - hierarchical [11] (see right)
    → **data serialization format**

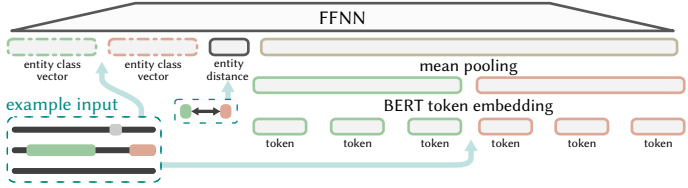
Note: all of the above evaluate on GPT models only.



Dunn, A. *et al. Structured information extraction from complex scientific text with fine-tuned large language models.* Dec. 2022. doi:10.48550/arXiv.2212.05238

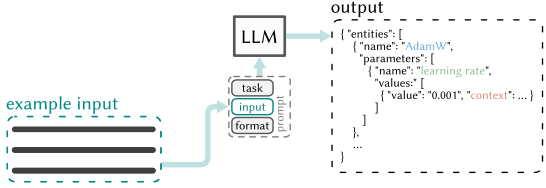
Approach: Fine-tuned models

- Based on PL-Marker (SciERC SOTA)
- (N)ER: used as is
- RE: new module, utilizing
  - Entity class embeddings
  - Entity distance



Approach: LLMs

- Data serialization format
  - JSON → YAML
- Compare 6 models
- Base prompt + tuning for each
- Zero-shot: all
- Few-shot: only Vicuna$_{16k}$



Approach: LLMs

- Data serialization format
  - JSON → YAML
- Compare 6 models
- Base prompt + tuning for each
- Zero-shot: all
- Few-shot: only Vicuna_{16k}

Model	Size
WizardLM [12]	13 B
Vicuna_{4k} [13]	13 B
Vicuna_{16k} [13]	13 B
Falcon [14]	40 B
GALACTICA [15]	120 B
GPT-3.5 [16]	175 B

Data

- No existing data sets
  → use unarXive [17] ML/CV/CL/DL
- Annotation approach
  - Paragraph level, whole papers
- Extent
  - 444 paragraphs
  - 1,971 entities
    (1,134 a, 131 p, 662 v, 44 c)
  - 614 relations
- IAA
  - 0.867 for entities
  - 0.737 for relations

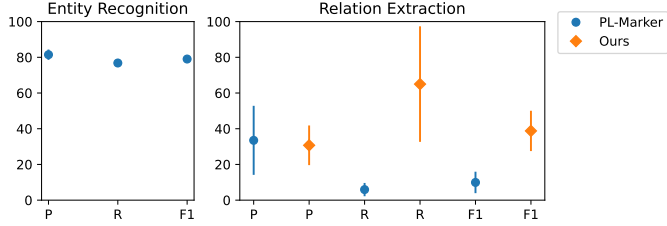


Experiments: Fine-tuned models

- **Setting**
  - 5-fold cross-validation
  - Stratified sampling
- **Results** (F_1 [%])
  - ER: 78.0
  - RE: 9.9 → 38.8
- **Analysis**
  - Parameter: low performance
  - Contexts: not predicted

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Table: Ablation study results (model inputs are: T = BERT token embeddings, C = entity class embeddings, D = entity distance)

Used	P [%]	R [%]	F_1 [%]
␣CD	15.5	8.8	11.1
T␣D	16.6	29.8	19.6
TC␣	26.5	65.0	35.5
TCD	**30.7**	**65.0**	**38.8**

Experiments: LLMs

- **Setting**
  - Zero-/5-shot
  - Compare JSON w/ YAML variant
- **Results** (best ER / RE F_1 [%])
  - Zero-shot: 37.4 / 7.8
  - 5-shot: 44.0 / 6.1
- **Analysis**
  - Very low RE performance
  - YAML: avg. +5% in ER
  - 5-shot: +27% ER, +6% RE
  - Format adherence
    + entity hallucinations

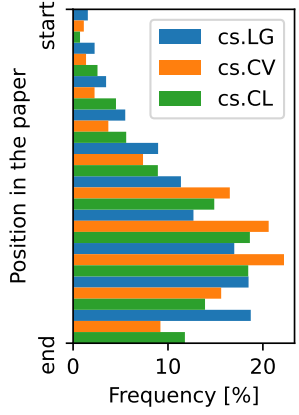
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    + entity hallucinations

Zero-shot		ER	RE
Model	Output	F_1 [%]	F_1 [%]
WizardLM	JSON	8.6	0.1
	YAML	15.3 $\Delta_{+6.7}$	0.1 $\Delta_{+0.0}$
Vicuna_{4k}	JSON	11.5	1.2
	YAML	22.3 $\Delta_{+10.8}$	0.1 $\Delta_{-1.1}$
Falcon	JSON	10.2	0.0
	YAML	19.8 $\Delta_{+9.6}$	0.0 $\Delta_{+0.0}$
GALACTICA	JSON	19.5	0.3
	YAML	21.1 $\Delta_{+1.6}$	0.1 $\Delta_{-0.2}$
GPT-3.5	JSON	33.8	7.2
	YAML	**37.4** $\Delta_{+3.6}$	**7.8** $\Delta_{+0.6}$
5-shot		ER	RE
Vicuna_{16k}	JSON	39.6	1.3
	YAML	**44.0** $\Delta_{+0.4}$	**6.1** $\Delta_{+4.8}$

Experiments: Application

- Apply best model (BERT based) on 15k paper sample
- Parameters information given in
  - 36% of ML papers
  - 42% of CV papers
  - 36% of CL papers
  - 7% of DL papers
- Distribution towards second half of paper accross disciplines



Conclusion

- Novel, relevant task
  (data scheme, annotation guidelines)
- High quality manually annotated data set
- Approaches based on BERT, LLMs
  - BERT model based approach
    - 29% F_1 increase for RE
  - LLM approach
    - Avg. 5.5% F_1 increase for ER
    - Consistent across all used LLMs
- Trained model applicable on large scale

Conclusion

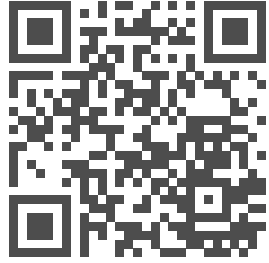
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    - Consistent across all used LLMs
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- **Future Work**
  - Larger data set (LLM fine-tuning)
  - More sophisticated/recent models
  - Annotation guidelines in prompt
    (large context LLMs)
  - Use LLMs for generating paragraphs from ground truth

Thank You ☺

Questions?

Code/Data/Paper



References I

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Extra Slides

Task

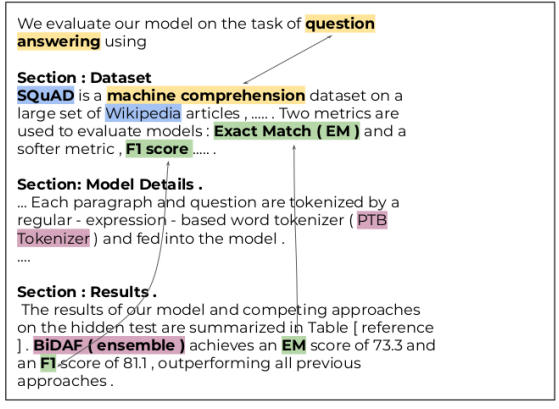
- **Task Type**
  - (Named) Entity Recognition, Relation Extraction
- 4 entity classes
  - (1) research **artifact**: model, method, data set, ...
  - descriptions of how authors use the artifacts
    (2) **parameter** (α, learning rate, k, ...)
    (3) **value** (1e-3, five, $\frac{1}{3}$, ...)
    (4) **context** (for fine-tuning, during grid search, ...)
- 1 relation type
  - Given by entity type pair

Task (ext)

- **Goal**
  - Automatically extract hyperparameter information from paper text
- **Motivation**
  - Reproducibility indication [19], automated reproduction [20]
  - Uncover conventions and trends
  - More fine-grained paper representations (similarity measures, recommendation, search)
- **Task Type**
  - (Named) Entity Recognition, Relation Extraction

Related Work: Fine-tuned models

- SciERC dataset
  - SCIIE [2]
  - PL-Marker [3]
  - → **entity type overlap**
- SciREX dataset
  - TempGen [4] (only RE)
- SemEval 2022 (math symb. to descr.)
  - JBNU-CCLab [5]
  - AIFB [6]
- SemEval 2021 (measurements)
  - LIORI [7]
  - → **utilize mention pattern regularities**



Related Work: Fine-tuned models: Other data sources

- From code documentation [21]
- From code [22]

Approach: LLMs

```
In the context of machine learning and related fields, what (if any) are the entities (datasets, models,
methods, loss functions, regularization techniques) mentioned in the LaTeX Input Text below? What (if any)
are their parameters and values?

[LaTeX Input Text start]
We use AdamW with a learning rate ($\alpha$) of 1e-3 for /* [...] */
[LaTeX Input Text end]

Answer in the following YAML format.

Format:
---
text_contains_entities: true/false
entities:
  - entity<N>:
      id: e<N>
      name: "<entity name>"
      type: dataset/model/method/loss function/regularization technique
      has_parameters: true/false
      parameters:
        - parameter<M>:
            id: p<N.M>
/* [...] */
...

Only include entities that are of type dataset, model, method, loss function, or regularization technique.
Do not output entities that are of another type. Do not include entities of type task, metric, library,
software, or API.
Only produce output in the YAML format specified above. Output no additional text.

Output:
```

Approach: LLMs

```
In the context of machine learning and related fields, what (if any) are the entities (datasets, models,
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```
Task

```
[LaTeX Input Text start]
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[LaTeX Input Text end]
```
Input Text

```
Answer in the following YAML format.

Format:
---
text_contains_entities: true/false
entities:
  - entity<N>:
      id: e<N>
      name: "<entity name>"
      type: dataset/model/method/loss function/regularization technique
      has_parameters: true/false
      parameters:
        - parameter<M>:
            id: p<N.M>
/* [...] */
...
```
Format

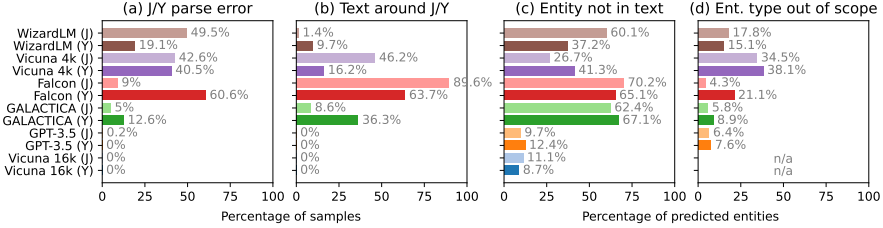
```
Only include entities that are of type dataset, model, method, loss function, or regularization technique.
Do not output entities that are of another type. Do not include entities of type task, metric, library,
software, or API.
Only produce output in the YAML format specified above. Output no additional text.

Output:
```
Output Prefix

Experiments: LLMs

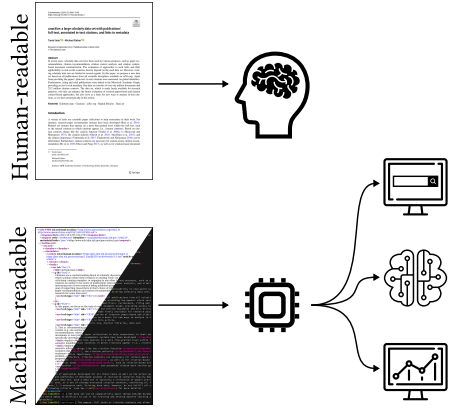
Zero-shot		Entity Recognition			Relation Extraction		
Model	Output	P [%]	R [%]	F_1 [%]	P [%]	R [%]	F_1 [%]
WizardLM	JSON	6.9	11.3	8.6	0.1	0.8	0.1
	YAML	9.7	35.6	15.3$_{\Delta+6.7}$	0.1	1.5	0.1$_{\Delta+0.0}$
Vicuna$_{4k}$	JSON	15.1	9.3	11.5	0.7	3.8	1.2
	YAML	17.3	31.5	22.3$_{\Delta+10.8}$	0.0	0.8	0.1$_{\Delta-1.1}$
Falcon	JSON	**37.1**	5.9	10.2	0.0	0.0	0.0
	YAML	32.7	14.2	19.8$_{\Delta+9.6}$	0.0	0.0	0.0$_{\Delta+0.0}$
GALACTICA	JSON	25.9	15.7	19.5	0.1	2.3	0.3
	YAML	23.1	19.5	21.1$_{\Delta+1.6}$	0.0	0.8	0.1$_{\Delta-0.2}$
GPT-3.5	JSON	27.9	**42.8**	33.8	5.4	10.7	7.2
	YAML	34.0	41.7	**37.4**$_{\Delta+3.6}$	**5.8**	**12.2**	**7.8**$_{\Delta+0.6}$
5-shot		Entity Recognition			Relation Extraction		
Vicuna$_{16k}$	JSON	34.4	46.7	39.6	0.8	4.6	1.3
	YAML	**43.9**	**44.1**	**44.0**$_{\Delta+0.4}$	**4.5**	**9.9**	**6.1**$_{\Delta+4.8}$

Experiments: LLMs

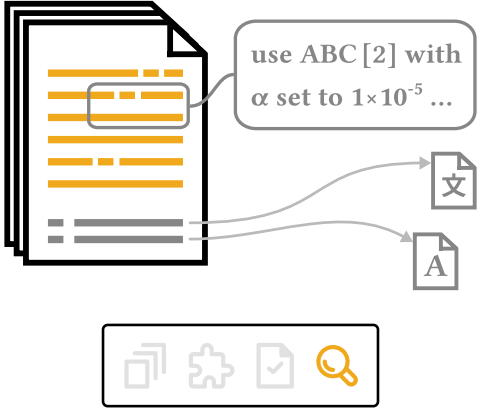


Scholarly Data Contexxt

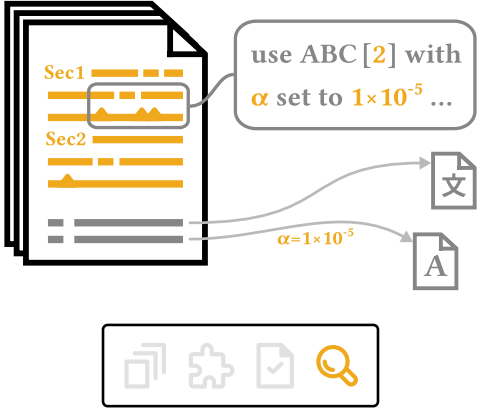
- **Usage**
  - Services (Search, Recommendation, Stats)
  - ML models (LLMs, sumarization, recommender systems)
  - Analyses (temporal, geographic, institutional)
- **Flavors**
  - Metadata (MAG, OpenAlex, ORKG, crossref)
  - Documents (Core, arXMLiv, PMC)
  - Linked Documents (unarXive, S2ORC)
- **Data Sources**
  - PDF (Core, ACL Anthology)
  - XML (PubMed, PLOS, publisher internal)
  - LaTeX (arXiv)



Citation Link Parameters



Citation Link Parameters



Overview

- **Research gap**
  - Important information not considered
  - Lack of link granularity
  - No formalization/approaches
- **Approach**
  - Task formalization
  - Data annotation (data from unarXive)
  - Two lines of approaches
    - BERT based model approach
    - LLM based approach
- **Results**
  - Novel task, novel data
  - Improvements over SOTA baselines

